

KNN and Decison Tree

Introduction To Binary Classification

Binary classification is a fundamental task in machine learning, commonly employed in various domains such as computer vision, natural language processing, and bioinformatics. Its objective is to assign objects into one of two categories based on their features.

Input

$$\{x_1, x_2, \dots, x_n\} \mid x_i \in \mathbb{R}^d$$

Label

$$\{y_1, y_2, \dots, y_n\} \mid y_i \in \mathbb{R}$$

Goal

$$h : \mathbb{R}^d \rightarrow \{0, 1\}$$

Zero-One Loss

$$\frac{1}{n} \sum_{i=1}^n \mathbb{1}(h(x_i) \neq y_i)$$

↓

$$\mathbb{1}(z) = \begin{cases} 1, & \text{if true} \\ 0, & \text{if false} \end{cases}$$

This is the average number of errors in prediction by the classifier

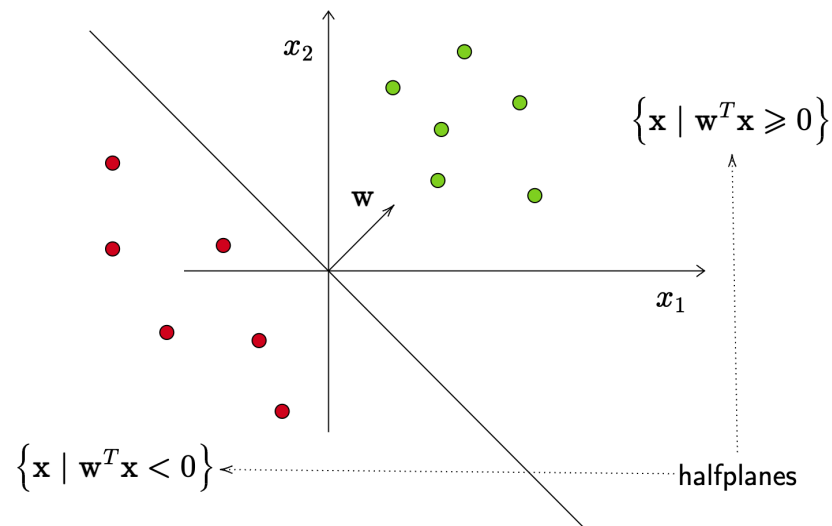
Linear Classifier

$$\min_{h \in \mathcal{H}_{linear}} \sum_{i=1}^n \mathbb{1}(h(x_i) \neq y_i)$$

$$\mathcal{H}_{linear} = \{h_w : h(x) = \text{sign}(w^T x)\}$$

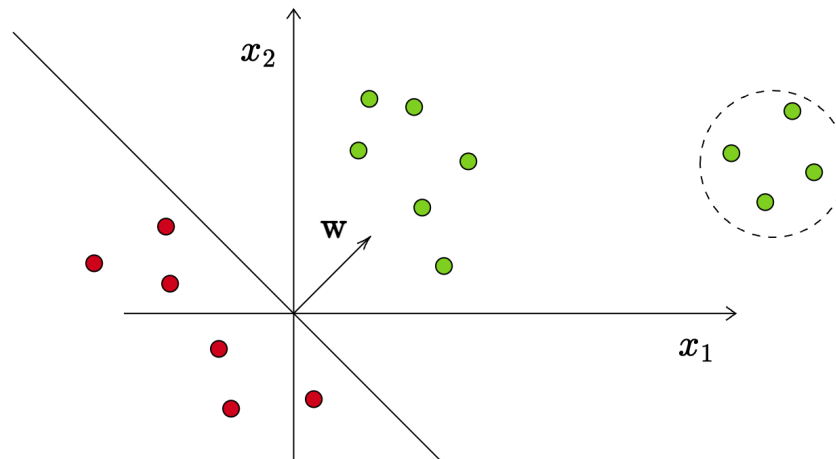
$$\text{sign}(z) = \begin{cases} 1, & \text{if } z > 0 \\ 0, & \text{otherwise} \end{cases}$$

The optimization problem given below is NP hard



Using Linear Regression To Solve Classification Problem ?

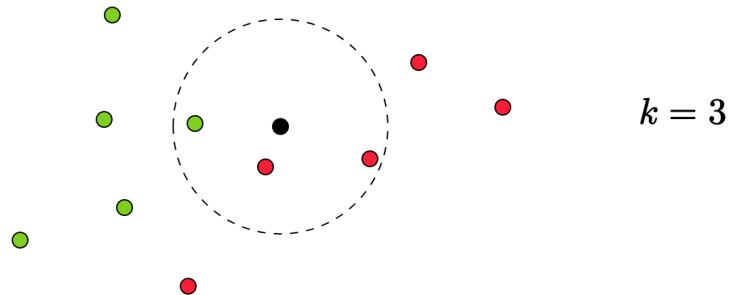
Using the SSE for classification is not a good idea. The SSE is sensitive to outliers, it becomes evident that linear regression-based classification not only separates the two categories based on their respective sides but also considers their positions. As a consequence, the classification boundary may shift with respect to the outliers in the dataset. Hence, this approach is not suitable for binary classification.



K-Nearest Neighbours Algorithm

The K-Nearest Neighbors (K-NN) algorithm is a widely-used non-parametric method utilized for both classification and regression tasks in machine learning. It operates by identifying the K-nearest data points to the target object and classifying or regressing the target object based on the majority of its nearest neighbors. The algorithm follows these steps:

- Given a test point $x_{test} \in \mathbb{R}^d$, find the k-closest points from x^* to x_{test} in the training set $\{x_1^*, x_2^*, \dots, x_n^*\}$
- Predict the label of the test sample as $y_{test} = \text{majority}(y_1^*, y_2^*, \dots, y_n^*)$



Issue ?

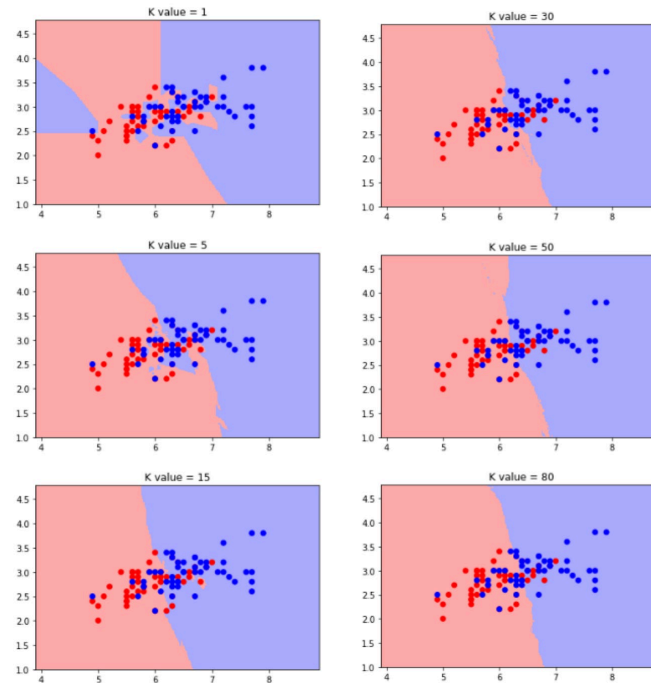
Can get affected by outliers (lets a negative datapoint among positive datapoint cluster)

Fix ?

Ask more neighbours before prediction

How Many Neighbours ?

As k increases, the model becomes less flexible, smaller the value of k , complicated the decision boundary.
Thus, Choosing k is a hyper-parameter, thus cross validate for k



If $K = 1$, the classification is highly sensitive to outliers leading to overfitting

If $K = n$, the classification becomes overly smooth leading to underfitting

Issues With KNN?

- The choice of distance function can yield different results. The Euclidean distance, commonly used, might not always be the best fit for all scenarios.
- Computationally, the algorithm can be demanding. When making predictions for a single test data point, the distances between that data point and all training points must be calculated and sorted.
- The algorithm does not learn a model but instead relies on the training dataset for making predictions.

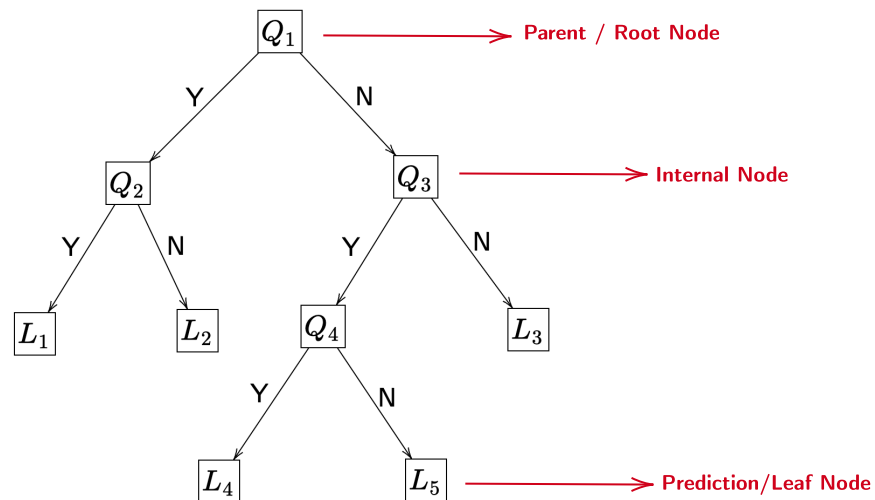
Introduction to Decision Tree

Decision trees are widely used in machine learning for classification and regression tasks. They operate by recursively partitioning the data based on the most informative features until a stopping criterion is met. Decision trees can be visualized as tree-like structures.

Input

$$\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}; x_i \in \mathbb{R}^d$$
$$y_i \in \{+1, -1\}$$

Output



Prediction

Given a datapoint x_{test} , traverse through the tree to reach a leaf node.

y_{test} = value in the leaf node

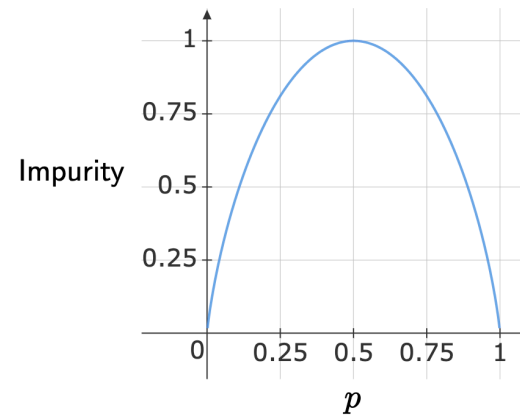
Question

Question is a (feature,value) pair. To measure the "Goodness of a Question", we need a measure of "impurity" for a set of labels $\{y_1, y_2, \dots, y_n\}$, we will use Entrophy function to measure the impurity.

$$\text{Entrophy}(p) = -(p \log p + (1-p) \log (1-p))$$

$$\text{here, } p = \frac{n_p}{n}$$

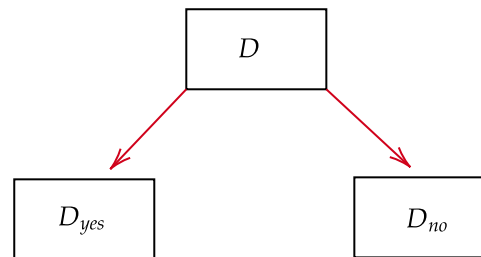
Pictorial representation of the Entrophy function



- Measure of impurity is highest when $p = \frac{1}{2}$
- Measure of impurity is 0 when $p = 0$ or 1

Information Gain is then utilized to measure the quality of a split in the decision tree algorithm.

Information gain is a commonly used criterion in decision tree algorithms that quantifies the reduction in entropy or impurity of a dataset after splitting based on a given feature. High information gain signifies features that effectively differentiate between the different classes of data and lead to accurate predictions. Information gain is calculated as



$$IG = E_D - [\gamma \text{entropy}(D_{yes}) + (1 - \gamma) \text{entropy}(D_{no})]$$
$$\text{here, } \gamma = \frac{|D_{yes}|}{|D|}$$

Decision Tree Algorithm

The decision tree algorithm follows these steps:

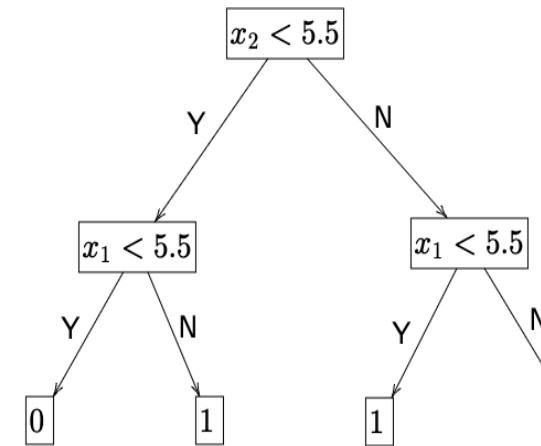
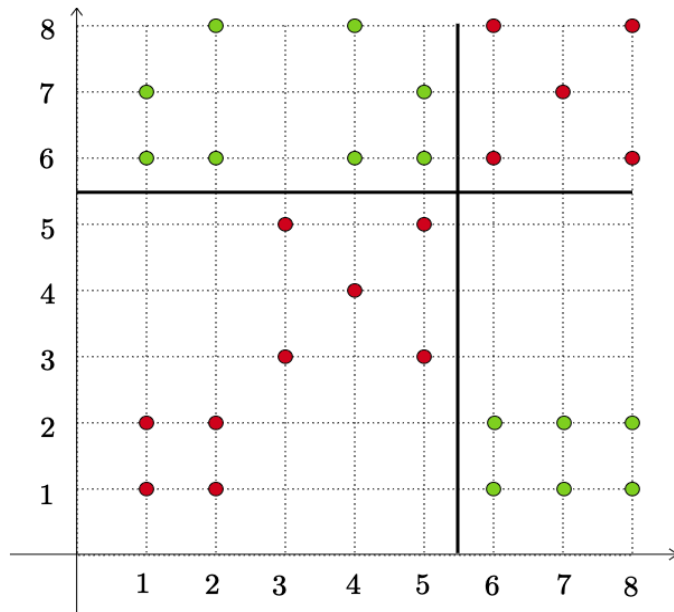
1. Discretize each feature within the range [min, max].
2. Select the question that provides the highest information gain.
3. Repeat the procedure for subsets D_{yes} and D_{no} .

Points to Remember

- Stop growing the tree when a node becomes sufficiently “pure”
- Depth of tree (The number of edges on the longest path from the root to a leaf) is a hyper parameter.
- Different measures, such as the Gini Index, can also be employed to evaluate the quality of a question.

Decision Boundary

In general, boundaries of the regions are parallel to the axes



Advantages and Disadvantages

Advantages

- Shallow trees are interpretable
- Easy to implement

Disadvantages

- Simple model with low predictive power.
- Deep trees have high variance, i.e; small changes to the training data result in big changes in the resulting tree.

Reference

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Karthik Thiagarajan, Machine Learning Techniques Github Notes.
Indian Institute of Technology Madras , CS2007 Course Website
Stanford University, CS229 Machine Learning Notes

Thank You !